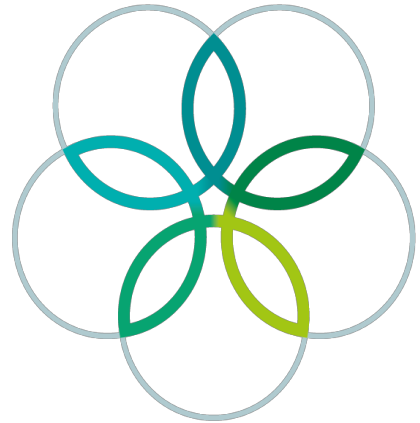
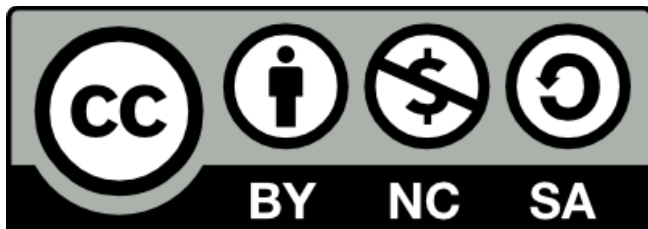


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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Student Code: _____

19th INTERNATIONAL BIOLOGY OLYMPIAD

13th – 20th July, 2008

Mumbai, INDIA



THEORETICAL TEST – PART B

ANSWER KEY FOR THE JURY

CELL BIOLOGY (26 points)

5. $2+1+2 = 5$
1. (~~2~~ $\times$ 3 = 6 points)

- a. Answer: $0.33 \times 10^{-8} \text{ M}$ — 2
b. Answer: $1.36 \times 10^{-3} \text{ m}$ — 1
c. Answer: $2.27 \times 10^{11} \text{ cells}$ — 2

2. ($0.5 \times 6 = 3$ points)

	Organ/Cell	SER extensively present	SER not extensively present	Function/s (if extensively present)
a.	Adrenal gland	✓		I
b.	Sebaceous glands	✓		I
c.	Intestinal villi	✓		I
d.	Muscles	✓		III
e.	Liver	✓		II and/or IV
f.	Pancreas		✓	

3. (0.5 x 4 = 2 points)

Situation I: A

Situation II: B

Situation III: B

Situation IV: A

4. (2 + 1 = 3 points)

a. Answer: 40 %

b. Answer: 1.5

5. (0.5 x 5 = 2.5 points)

1	2	3	4	5
$\frac{E}{A}$	$\frac{A}{E}$	B	C	D

6. (0.5 x 8 = 4 points)

I. 5.6

II. 6.3

III. 0.5

IV. 0.21

V.

	True	False
a.	√	
b.		√
c.		√
d.	√	

7. (1 x 3 = 3 points)

I.

a.	b.	c.	d.
		√	

II.

a.	b.	c.	d.
√			

III.

a.	b.	c.	d.
	√		

8. (0.5 x 5 = 2.5 points)

Protein	Mode of regulation			
	I	II	III	IV
A		√		
B				√
C	√			
D		√	√	

9. (0.5 x 8 = 4 points)

No.		Answer
I	Cell/s that is/are not alive when functional.	A, B, F
II	Plasmodesmata can be found associated with this/these cell/s.	C, D, E
III	When you eat potato, you eat the tissue formed of this/these cell/s.	D
IV	Cell/s that harden/s the nut skin.	F

✓ 10. (0.5 x 3 = 1.5 points)

Graph	Plant type
A	II
B	III
C	I

✓ 11. (0.5 x 4 = 2 points)
(A)

Region	Water potential
P	- 1 atm
Q	- 5 atm
R	- 8 atm

✓ (B)

a.	b.	c.	d.
	✓		

12. (1 for each row x 4 = 4 points)

	<i>Chlamydomonas</i>	Cyano- bacteria	Green- sulphur bacteria	Purple- sulphur bacteria
Phototrophic autotrophs	✓	✓	✓	✓
Photosystem II absent			✓	✓
Respiratory enzymes located on plasma membrane		✓	✓	✓
Chlorophyll <i>a</i> as the major photosynthetic pigment	✓	✓		

13. (0.5 x 7 = 3.5 points)

Process	+ / –
1	–
2	+
3	+
4	–
5	+
6	+
7	–

✓ 14. (2 points)

a.	b.	c.	d.
		✓	

✓ 15. (0.5 x 4 = 2 points)

	P	Q	R	S
Liver				✓
Brain	✓			
Thymus			✓	
Gonads		✓		

✓ 16. (0.5 x 4 = 2 points)

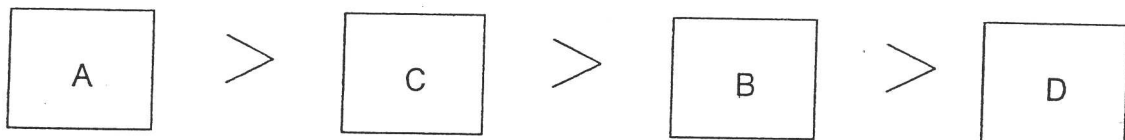
	True	False
a.		✓
b.		✓
c.	✓	
d.	✓	

✓ 17. (0.5 x 4 = 2 points)

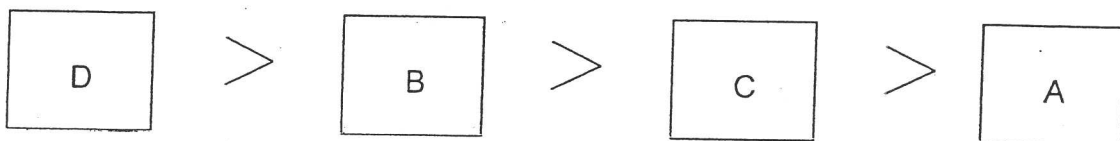
Set	Condition	True	False
I	Curve I. Normal blood pH and Curve II. Acidosis	✓	
II	Curve I. 40°C and Curve II. 30°C		✓
III	Curve I. Elephant hemoglobin and Curve II. Cat hemoglobin	✓	
IV	Curve I. Fetal hemoglobin and Curve II. Maternal hemoglobin	✓	

✓ 18. (1 x 2 = 2 points)

Surface area per unit volume of the body



Total volume of blood in the body



✓ 19. (1 + 1 + 0.5 x 6 = 5 points)

a.

a.	b.	c.	d.
	✓		

✓ b.

a.	b.	c.	d.
		✓	

✓ c.

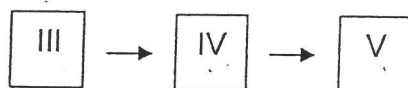
Options	True	False
a.		✓
b.	✓	
c.		✓
d.	✓	
e.		✓
f.		✓

✗ 20. (0.5 x 6 = ^{3.5}3 points)

A.



B.



21. (2 points)

Answer: 27/64 or 0.4219

✓ 22. (0.5 x 4 = 2 points)

	I	II	III
Lactose hydrolysis by β -galactosidase			✓
Reduction of <i>lac</i> repressor's affinity for the <i>lac</i> operator		✓	
Binding of the CAP-cAMP complex to the <i>lac</i> promoter		✓	
Utilization of glucose	✓		

23. (2 points)

Answer: 43.52 %

24. (2 points)

Answer: 1/6 or 0.1667 , 1.7

25. (2 points)

Answer: 9

✓ 26. (1 x 2 = 2 points)

a. /

Yes	No
✓	

✓ b. Answer: 0.24

27. (1 x 2 = 2 points)

a. Answer: 0.5192

b. Answer: 0.3696

~~28. (2 points)~~

Answer: 1/10 or 0.1

~~29. (1 x 2 = 2 points)~~

I.

a.	b.	c.	d.
		✓	

II.

a.	b.	c.	d.
	✓		

18.6 kJ

30. (1 + 0.5 x 3 = 2.5 points)

(A)

p	
q	
r	$\sqrt{\quad}$

(B)

Distance between p and q	28.5 mu
Distance between p and r	17.5 mu
Distance between q and r	11 mu

31. (0.5 x 3 = 1.5 points)

Growth curve

Survivorship Curve

Age Structure

P	Q	R	S	T	U
$\sqrt{\quad}$			$\sqrt{\quad}$	$\sqrt{\quad}$	

GC	SC	AS
$\sqrt{\quad}$	$\sqrt{\quad}$	$\sqrt{\quad}$

32. (1 x 3 = 3 points)

(A)

a.	b.	c.	d.
	$\sqrt{\quad}$		

✓ (B)

a.	b.	c.	d.
	✓		

✓ (C)

a.	b.	c.	d.
		✓	X

33. (0.5 x 4 = 2 points)

Number	A	B	Type of interaction
1.	+	0	II
2.	+	+	IV
3.	+	–	V
4.	+	+	III

✓ 34. (1 x 4 = 4 points)

(A)

a.	b.	c.	d.
✓			

✓ (B)

a.	b.	c.	d.
		✓	

✓ (C)

a.	b.	c.	d.
✓			

✓ (D)

a.	b.	c.	d.
✓			

35. $(0.5 + 0.5 + 0.5 + 1 + 0.5 + 0.5 + 0.5 + 0.5 + 1 + 0.5 = 6 \text{ points})$

(A)

I. ✓

a.	b.	c.	d.
		✓	

II. ✓

a.	b.	c.	d.
✓			

III. Answer: 8

IV. Answer: 0.72

V.

a.	$\sqrt{\quad}$
b.	

(B)

I.

a.	b.	c.	d.
		$\sqrt{\quad}$	

II.

a.	b.	c.	d.
	$\sqrt{\quad}$		

III. Answer: 1

IV. Answer: 1.82

1.8, 1.8

V.

a.	$\sqrt{\quad}$
b.	

36. (2 points)

(A)

a.	
b.	✓

(B)

a.	b.	c.	d.
		✓	

37. (0.5 x 6 = 3 points)

(A)

		Opponent	
		Hawk	Dove
Attacker	Hawk	-25	+50
	Dove	0	+15

(B)

Statement	True	False
a.		✓
b.		✓

38. (0.5 x 4 = 2 points)

Physiological change <i>Behavioral change</i>	Option/s
A	IV and/or I
B	III
C	II
D	IV and/or I

Physiological changes.

39. (2 x 2 = 4 points)

(A)

a.	b.	c.	d.
		✓	

(B)

a.	b.	c.	d.
			✓

40. (2 points)

Taxon	Option
T3	VII
T2a	VIII or X or VI, respectively
T1a	XVI or XV or XIII, respectively

T2b	VIII or X or VI, respectively
T1b	XVI or XV or XIII, respectively
T2c	VIII or X or VI, respectively
T1c	XVI or XV or XIII, respectively

41. (2 points)

a.	b.	c.	d.
√			

42. (0.5 x 10 = 5 points)

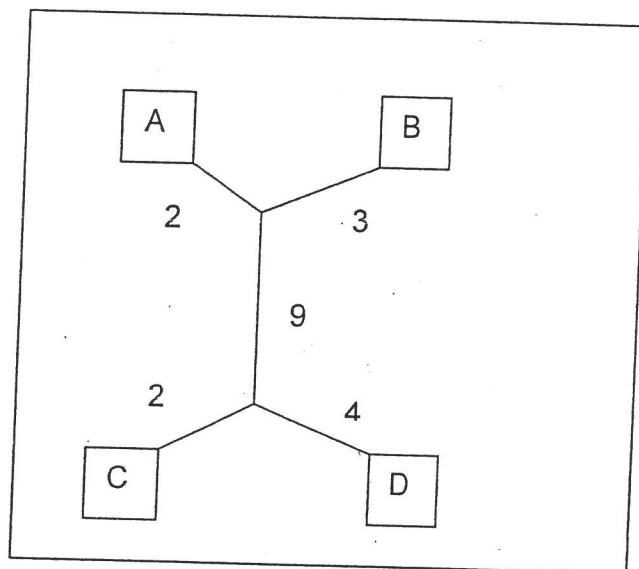
Group	Number	Group	Number
Annelida (Earthworms)	2	Mollusca (Snails)	5
Arthropoda (Crayfishes)	3	Mollusca (Squids)	1
Cnidaria (Jellyfishes)	7	Nematoda (Roundworms)	9
Echinodermata (Starfishes)	6	Platyhelminthes (Tapeworms)	10
Mollusca (Bivalvia)	4	Porifera (Sponges)	8

43. (1 + 3 = 4 points)

(A)

a.	b.	c.	d.
		✓	

(B)



***** END OF PART B *****